

Name: _____

1. Code Tracing

Consider the following recursive function:

```
def mystery(n):  
    if n == 1:  
        print("Base case")  
        return 1  
  
    print("Calling mystery with", n - 1)  
    result = mystery(n - 1)  
    print("Returning from", n)  
    return result + n
```

What will be printed when the following code is executed?

```
answer = mystery(4)  
print("Answer:", answer)
```

Write the complete output below.

2. Writing a Recursive Function

Write a recursive function called `product_to_n` that takes a positive integer `n` and returns the product of all integers from 1 to `n`.

Read that again.

For example:

```
print(product_to_n(4))
```

should print:

```
24
```

Your function must use recursion and **MUST NOT USE A LOOP**. Write clear indentation.

```
def product_to_n(n):  
    # Write your code below each comment  
    # BASE CASE:  
  
    if _____:  
        return _____  
  
    # RECURSIVE CASE:  
  
    return _____
```

3. Writing a Recursive Function

Write a recursive function called `count_letter` that takes a string and a letter and returns the number of times that letter appears in the string.

For example:

```
print(count_letter("banana", "a"))
```

should print:

```
3
```

Your function **must use recursion** and must NOT use a loop or the count method. Careful with indentation.

```
def count_letter(word, letter):  
    # Write your code below each comment
```